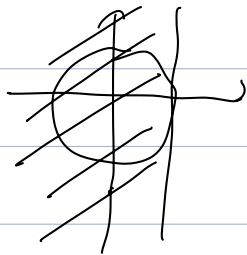


Direct numerical integration

forward rectangular: $s \in \frac{z-1}{T}$

$$\Rightarrow z = sT + 1 = 1 + j\omega T$$



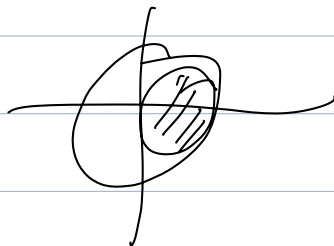
backward rectangular $s \in \frac{z-1}{Tz}$

$$\Rightarrow z = \frac{1}{1 - Ts} = \frac{1}{1 - j\omega T}$$

$$= \frac{1}{2} + \frac{1}{1 - j\omega T} - \frac{1}{2}$$

$$= \frac{1}{2} + \frac{1 + j\omega T}{2(1 - j\omega T)}$$

$$\Rightarrow \left| z - \frac{1}{2} \right| = \left| \frac{1}{2} \cdot \frac{1 + j\omega T}{1 - j\omega T} \right| = \frac{1}{2}$$



Bilinear: $s \in \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$

$$\Rightarrow (1 + z^{-1})s = \frac{2}{T} (1 - z^{-1})$$

$$\Rightarrow z^{-1} \left(s + \frac{2}{T} \right) = \frac{2}{T} - s$$

$$\Rightarrow Z^{-1} = \frac{\frac{K}{T} \cdot s}{\frac{K}{T} + s}$$

$$\Rightarrow Z = \frac{1 + \frac{T}{2}s}{1 - \frac{T}{2}s} = \frac{1 + \frac{T}{2}j\omega}{1 - \frac{T}{2}j\omega}$$

$$|Z|: 1$$
